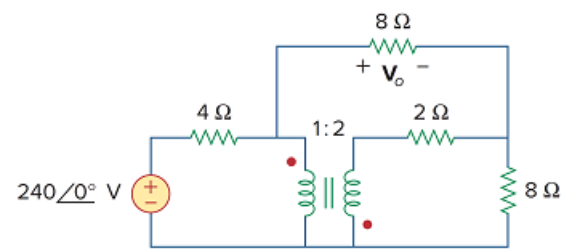


Problem

Find V_o in the circuit of Fig. 13.40.

Figure 13.40



Step-by-step solution

Step 1 of 12

Refer to Figure 13.40 from the textbook. Assume the directions of the loop currents, I_1 , I_2 and I_3 as shown in Figure 1.

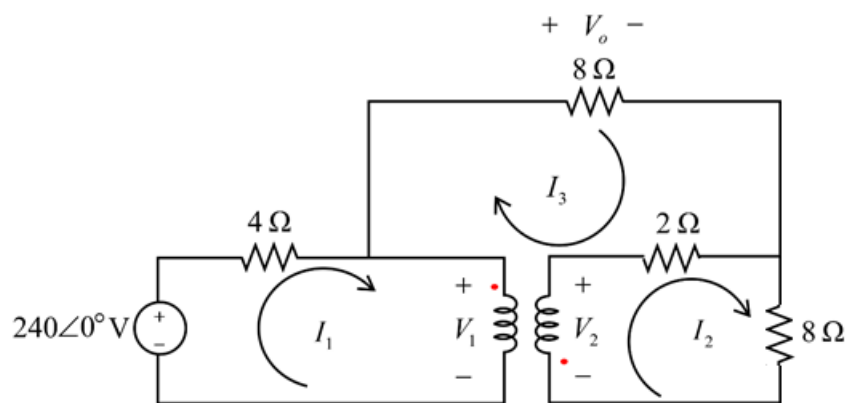


Figure 1

Step 2 of 12

Use Ohm's Law to write the expression for the voltage, V_o .

$$V_o = 8I_3 \dots\dots (1)$$

Step 3 of 12

Apply Kirchhoff's Voltage Law in the loop with loop current I_1 .

$$240 = 4I_1 + V_1 \dots\dots (2)$$

Apply Kirchhoff's Voltage Law in the loop with loop current I_2 .

$$V_2 = 2(I_2 - I_3) + 8I_2$$

$$V_2 = 10I_2 - 2I_3$$

$$0 = 10I_2 - 2I_3 - V_2 \dots\dots (3)$$

Apply Kirchhoff's Voltage Law in the loop with loop current I_3 .

$$0 = 2(I_3 - I_2) + 8I_3$$

$$0 = 10I_3 - 2I_2 + V_2 - V_1$$

$$0 = -2I_2 + 10I_3 - V_1 + V_2 \dots\dots (4)$$

Step 4 of 12

Write the voltage ratio for the transformer.

$$\frac{V_1}{V_2} = -\frac{N_1}{N_2}$$

Negative sign is due to the opposite polarity of the voltages.

Substitute $\frac{1}{2}$ for $\frac{N_1}{N_2}$.

$$\frac{V_1}{V_2} = -\frac{1}{2}$$

$$V_2 = -2V_1$$

Step 5 of 12

Substitute $-2V_1$ for V_2 in equation (3).

$$0 = 10I_2 - 2I_3 - (-2V_1)$$

$$0 = 10I_2 - 2I_3 + 2V_1 \dots\dots (5)$$

Step 6 of 12

Substitute $-2V_1$ for V_2 in equation (4).

$$0 = -2I_2 + 10I_3 - V_1 + (-2V_1)$$

$$0 = -2I_2 + 10I_3 - 3V_1$$

$$V_1 = \frac{-2I_2 + 10I_3}{3}$$

Step 7 of 12

Substitute $\frac{-2I_2 + 10I_3}{3}$ for V_1 in equation (2).

$$240 = 4I_1 + \left(\frac{-2I_2 + 10I_3}{3} \right)$$

$$720 = 12I_1 - 2I_2 + 10I_3 \dots\dots (6)$$

Step 8 of 12

Substitute $\frac{-2I_2+10I_3}{3}$ for V_1 in equation (5).

$$0 = 10I_2 - 2I_3 + 2\left(\frac{-2I_2+10I_3}{3}\right)$$

$$0 = 30I_2 - 6I_3 - 4I_2 + 20I_3$$

$$0 = 26I_2 + 14I_3$$

$$I_2 = -\frac{14}{26}I_3$$

$$= -\frac{7}{13}I_3$$

Step 9 of 12

Write the current ratio of the transformer.

$$\frac{I_S}{I_P} = -\frac{N_1}{N_2} \dots\dots (7)$$

Here,

I_P is the current in primary winding.

I_S is the current in secondary winding.

Thus,

$$I_P = I_1 - I_3$$

$$I_S = I_2 - I_3$$

Step 10 of 12

Substitute $I_1 - I_3$ for I_P and $I_2 - I_3$ for I_S in equation (7).

$$\frac{I_2 - I_3}{I_1 - I_3} = -\frac{N_1}{N_2}$$

Substitute $\frac{1}{2}$ for $\frac{N_1}{N_2}$.

$$\frac{I_2 - I_3}{I_1 - I_3} = -\frac{1}{2}$$

$$I_1 - I_3 = -2(I_2 - I_3)$$

$$I_1 + 2I_2 - 3I_3 = 0$$

Substitute $-\frac{7}{13}I_3$ for I_2 .

$$I_1 + 2\left(-\frac{7}{13}I_3\right) - 3I_3 = 0$$

$$I_1 - \left(\frac{14}{13} + 3\right)I_3 = 0$$

$$I_1 - \frac{53}{13}I_3 = 0$$

$$I_1 = \frac{53}{13}I_3$$

Step 11 of 12

Substitute $\frac{53}{13}I_3$ for I_1 and $-\frac{7}{13}I_3$ for I_2 in equation (6).

$$720 = 12\left(\frac{53}{13}I_3\right) - 2\left(-\frac{7}{13}I_3\right) + 10I_3$$

$$720 = \left(\frac{636}{13} + \frac{14}{13} + 10\right)I_3$$

$$720 = \left(\frac{636 + 14 + 130}{13}\right)I_3$$

$$I_3 = \left(\frac{720}{780}\right)(13)$$

$$I_3 = 12 \text{ A}$$

Step 12 of 12

Substitute 12 A for I_3 in equation (1).

$$\begin{aligned} V_o &= 8(12) \\ &= 96 \text{ V} \end{aligned}$$

Therefore, the voltage, V_o is 96 V.